

Ajay Venkatesh

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EDUCATION

New York University | Master of Science in Computer Engineering (GPA – 3.9/4)

Graduation - May 2026

Visvesvaraya Technological University | Bachelors in ECE (GPA - 8.9/10)

Graduation - Aug 2021

SKILLS

Programming languages: C, C++, JavaScript (ES6+ features), TypeScript, Python, Java, SQL.

Web Development: HTML, CSS, React, Node.js, Bootstrap, Express.js, Sequelize.

Database: RDBMS (MySQL, PostgreSQL), NoSQL (MongoDB, JSON), Dataverse.

Cloud: AWS (EC2, S3, Amplify, Lambda Function, API Gateway, Cognito, IAM, QuickSight, Q, SageMaker, CloudWatch, CloudFront), Azure (AI services, Logic Apps, Databricks, Function Apps), GCP (Firestore, BigQuery, Cloud Functions).

Tools: Git, Azure DevOps, Visual Studio, Jupiter, Dynamics CRM, ERP, Power Automate, Power BI, QuickSight.

Methodology: Agile, Waterfall.

AI/ML: Classification, Pattern Recognition, Data Extraction, MLOps, Neural Network, PCA, LLM.

EXPERIENCE

New York University, NYC, US

Aug 2024 - Present

Software Developer (On-Campus)

- Worked on an AI-driven Remote Patient Monitoring System, funded by the National Science Foundation, USA, employing React Native, TypeScript, and Cognito authentication for real-time health monitoring.
- Performed an ETL process to convert unstructured NoSQL JSON data into structured SQL data using GCP cloud functions. built visualization dashboard in AWS Quick sight which improves data analytics by 30%.
- Developed and deployed a React website with an embedded QuickSight dashboard, hosted using AWS S3 and CloudFront. Utilized Cognito, Amplify, and Lambda functions to establish login, role-based access control, and backend functionality.

Price Water House Coopers, (PwC)

Jul 2023 - Aug 2024

Senior Software Engineering

- Analyzed business goals and client requirements to synthesize a comprehensive product roadmap and solution architecture for Cloud based Health Application.
- Constructed a Cognitive Neural model to extract data from cards, integrated via API, reducing manual data processing time by 75%.
- Mentored 5 interns, conducted thorough code reviews for a team reducing bug occurrences by 25% and enabling a faster deployment cycle, aligning with agile methodologies.
- Engineered and optimized serverless database structure that improved data storage, retrieval, and processing workflows; achieved a 20% reduction in latency across diverse digital environments for seamless operations.
- Implemented CI/CD pipelines to streamline the solution migration to 3 higher environments, resulting in a 90% decrease in deployment time and enhanced software development lifecycle.

Associate Software Engineering

Aug 2021 - Jun 2023

- Designed 6 targeted security protocols that ensured appropriate control over logged-in users’ privileges by assessing role hierarchies: fortifying data protection measures while eliminating virtually all instances of unauthorized access.
- Built a B2B responsive, user-friendly web-based application for handling acquisition, creating deals, and onboarding, with capabilities like external data importing, data visualization using Power BI and Email-tracking.
- Optimized an automation flow used to clean, map and import 100K rows of data from Excel file to Database from 5 hours to 1 hour, resulting in an 80% increase in efficiency.
- Composed client and server-side custom configurations using JavaScript, plugins, custom workflows, webhooks, Business rules yielding a 35% faster server response.

PROJECTS

E-commerce store (Full Stack) | HTML, CSS, JavaScript, React, Mongo DB, Node JS, AWS (amplify, lambda, S3)

- Created a React JS shopping website, securing user data with AWS Cognito and processed payments via Lambda functions, utilized DynamoDB for data storage, and used AWS S3 for multimedia content.

Custom CSS Library (frontend library) | HTML, CSS, JavaScript

- Engineered a responsive CSS library with 10+ bootstrap-inspired components, featuring click-to-add functionality, easy integration, customization, and adaptive design for diverse devices. assists in rapid UI development reducing UI development time by 50%.

Hospital Management System (Full Stack) | React, TypeScript, CSS, SQL, Node JS, AWS (S3)

- Designed a React application to manage Appointments, Patient and Doctor records, Lab Reports and Medical History.
- Implemented MySQL data consistency measures via triggers and stored procedures, reducing anomalies by 50% and enhancing performance by 40%.

CERTIFICATIONS

AWS Cloud Practitioner

Microsoft Solution Architect Expert

Microsoft Developer Associate

Azure Data Scientist Associate

Power BI Data Analyst Associate

Azure Data Fundamentals

PUBLICATIONS

Supervised Machine Learning System Based Segmentation and Classification of Stroke Using Deep Learning Techniques [\(IEEE\)](#)

An Aquila-optimized SVM classifier for Diabetes prediction [\(IEEE\)](#)

TRAININGS

Complete Web Development Bootcamp »

Jan 2021 - June 2021

Data Structures and Algorithms (NPTEL- IIT Kanpur) »

July 2019 - Sep 2019